

System interconnect architecture

The system interconnect between computer subsystems and for constructing multiprocessors or multi-computers we can use.

- Static Networks
- Dynamic Networks

These networks can be used for internal connections among processors, memory modules, and I/O adaptors in a centralized system.

Network properties and routing:

- The topology of interconnect network can be either static or dynamic.
- Static networks are formed of point to point direct connections.
- Static networks are used for fixed connections among subsystems of a centralized system or multiple computing nodes of a distributed network.
- Dynamic networks are implemented with switched channels and they are dynamically configured to match the communication demand in user programs.
- Dynamic networks include buses, crossbar switches, multistage networks and routers.
- Packet switching plays an important role in modern multi-processor architecture.

Node Degree and Network Diameter:

- The number of edges (links or channels) incident on a node is called the node degree. It is denoted by d .
- The number of channels into a node is the in degree, and that out of a node is the out degree.
- Final node degree is the sum of the two in node degree and out node degree.
- Increasing node degree may cause increase in the cost of a node.
- The diameter D of a network is the smallest path between any two nodes.
- The path length is measured by the number of links traversed.
- The network diameter indicates the maximum number of distinct hops between any two nodes.
- Thus the network diameter should be as small as possible from a communication point of view.

Bisection width:

- When a given network is cut into two equal halves, the minimum number of edges along the cut is called the **channel BISECTION WIDTH b** .
- In the case of a communication network, each edge may correspond to a channel with w bit wires.
- Then the **wire bisection width** is $B = \frac{b}{w}$.
- The parameter B reflects the wiring density of a network and when B is fixed, **the channel width (in bits) $w = \frac{B}{b}$** .
- Another relevant parameter is the wire length or channel length between nodes. This parameter may affect the signal latency, clock skewing or power requirements.

Data routing functions:

- A data-routing network is used for inter PE(processing element) data exchange.
- This routing network can be static, such as the hypercube routing network used in the TMC/CM-2 or dynamic such as multistage network used in the IBM GF11.

*The Connection Machine 2 (CM-2) is an example of a practical large scale SIMD supercomputer.

*IBM GF11 was a parallel computer designed for the numerical solution of problems in **quantum chromodynamics (QCD)** and based on the model of parallel computing.

- In case of multi-computers, the data routing is achieved by the message passing where hardware routers are used to route messages among multiple computer nodes.
- The routing functions among processing elements include shifting, rotation, permutation (one to one), broadcast (one to all), multicast (one to many), shuffle, exchange, etc.
- These routing functions can be implemented on a ring, mesh, hypercube or multistage networks.

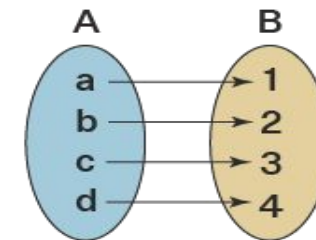
PERMUTATIONS:

- It is a type of data routing function (one to one).
- For n objects, there are $n!$ ways in which they can be reordered.
- Permutations can be implemented using crossbar switches, multistage networks, shifting, and broadcast operations.
- The time required to perform permutations of the connections between nodes often dominates the network performance when n is large.
- Eg. : the permutation $\pi = (a,b,c)(d,e)$ has
- bijection mapping $a \mapsto b, b \mapsto c, c \mapsto a, d \mapsto e$
- and $e \mapsto d$ in a circular fashion.

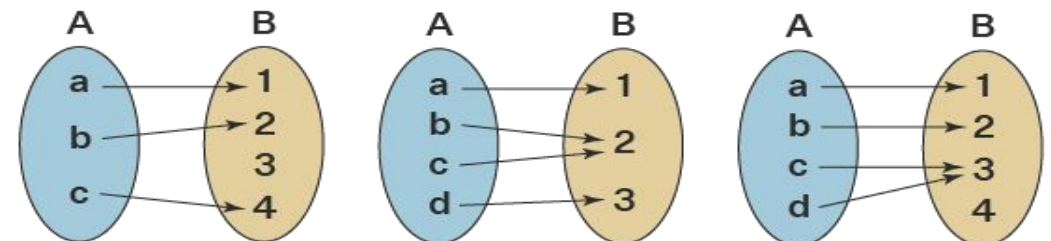
Bijjective Function - One-to-One & Onto Function



Bijjective Function



Not a Bijjective Function

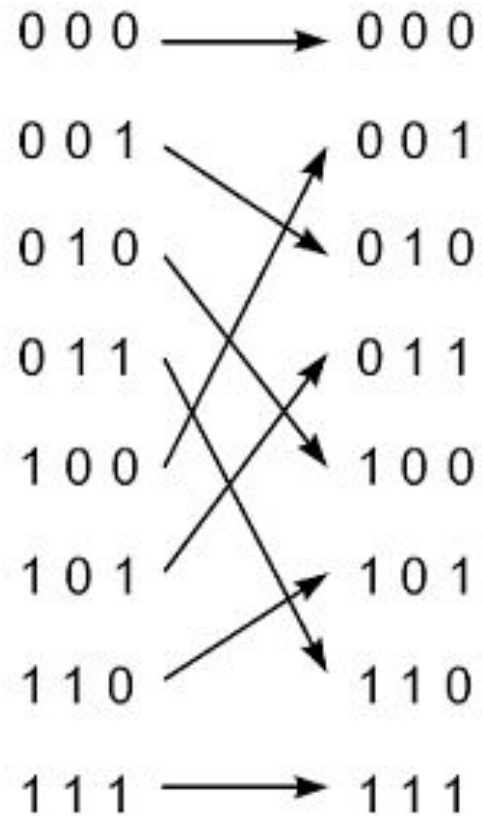


Applications of Permutations

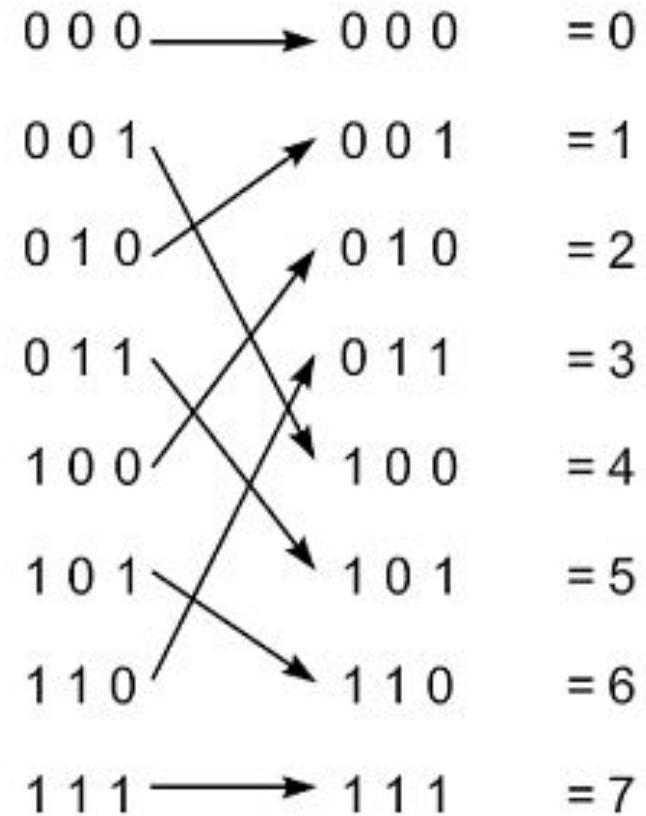
- **Matrix Computation:** Permutations allow conflict-free access to rows, columns, diagonals, and square blocks of matrices.
- **Parallel Processing:** They facilitate efficient data exchange between processors and memory in distributed systems.
- **Shifting/Sorting:** They are used to rearrange data words, often implemented as "permute instructions" in processors for tasks like multimedia, graphics, or cryptography

Perfect shuffle

- It is a special permutation function suggested by Harold Stone for parallel processing applications.
- This is shifting 1 bit to the left and wrapping it around to the least significant bit position (fig. a)
- The inverse perfect shuffle reverses the effect of the perfect shuffle. (fig. b)



(a) Perfect shuffle



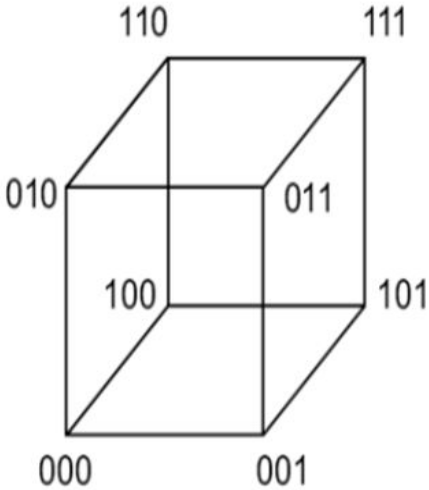
(b) Inverse perfect shuffle

Hypercube Routing Functions

- Hypercube or binary n-cube multiprocessor structure is composed of $N=2^n$ processors interconnected in an **n** dimensional binary cube. (N is no. of processors)
- one bit differs between each pair of adjacent vertices.
- For example, with a 3-dimensional cube, we can easily identify routing functions that exchange data between nodes with addresses that differ in the least significant, most significant, or middle bit.

Each processor in hypercube is linked to $\log_2 n$ adjacent nodes in a n cube network.

For eg. if $n=8$, the no. of adjacent processors linked is 3

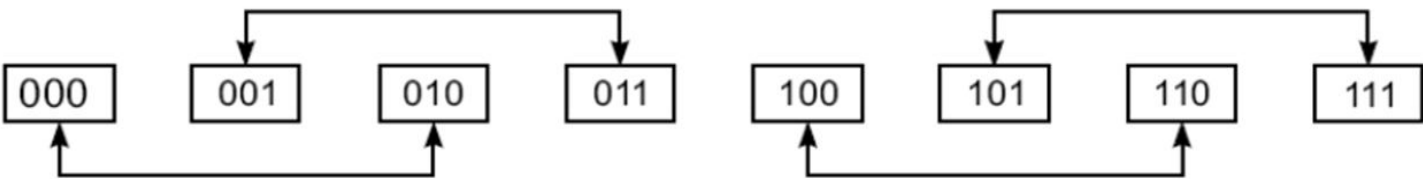


(a) A 3-cube with nodes denoted as $C_2C_1C_0$ in binary

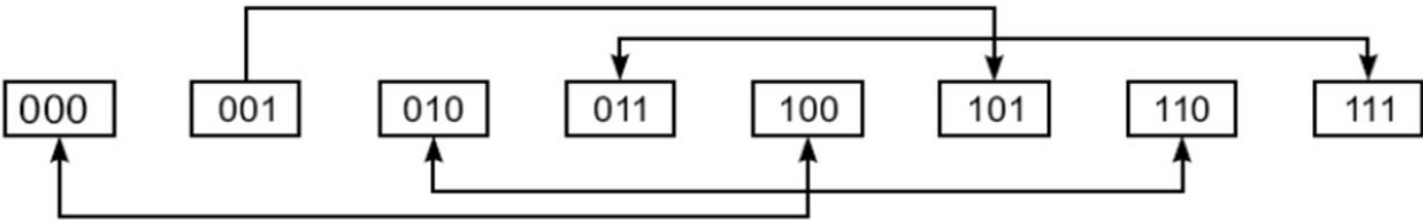
(a) A 3-cube with nodes denoted as $C_2C_1C_0$ in binary



(b) Routing by least significant bit, C_0



(c) Routing by middle bit, C_1



(d) Routing by most significant bit, C_2

Broadcast and multicast:

- Broadcast is a one-to-all mapping. For eg., in an SIMD computer using a broadcast bus extending from the array controller to all PEs. A message passing multicomputer also has mechanisms to broadcast messages.
- Broadcast is often treated as a global operation in a multicomputer. Multicast has to be implemented with matching of destination codes in the network.

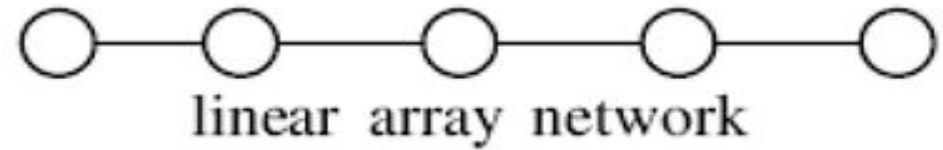
Network performance:

The performance of an interconnection network is affected by the following factors:

- 1. Functionality:** this refers to how network supports data routing, interrupt handling, synchronization, request/message combining and coherence.
- 2. Network latency:** this refers to worst case time delay for a unit message to be transferred through the network.
- 3. Bandwidth:** the maximum data transfer rate, in terms of Mbps or Gbps transmitted through the network.
- 4. Hardware complexity:** refers to the implementation costs such as wires, switches, connectors, arbitration and interface logic.
- 5. Scalability:** refers to the ability of a network to be modularly expandable with a scalable performance with increasing machine resources.

Static connection network

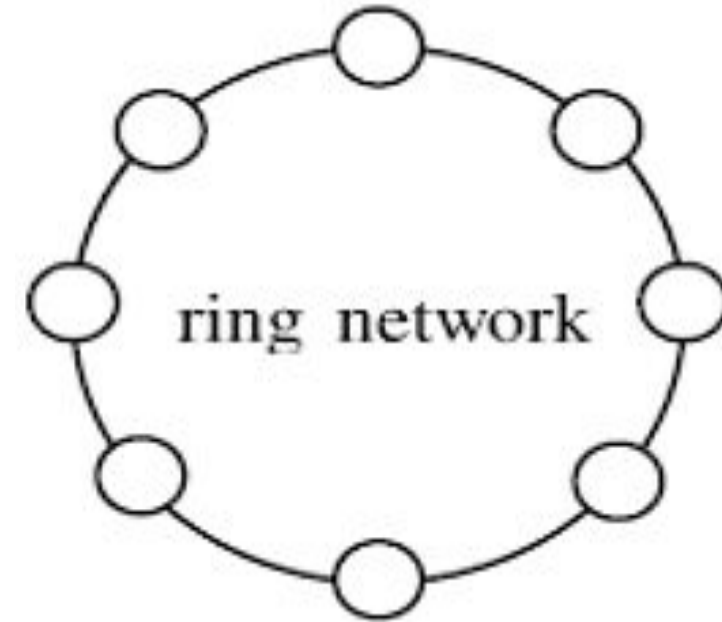
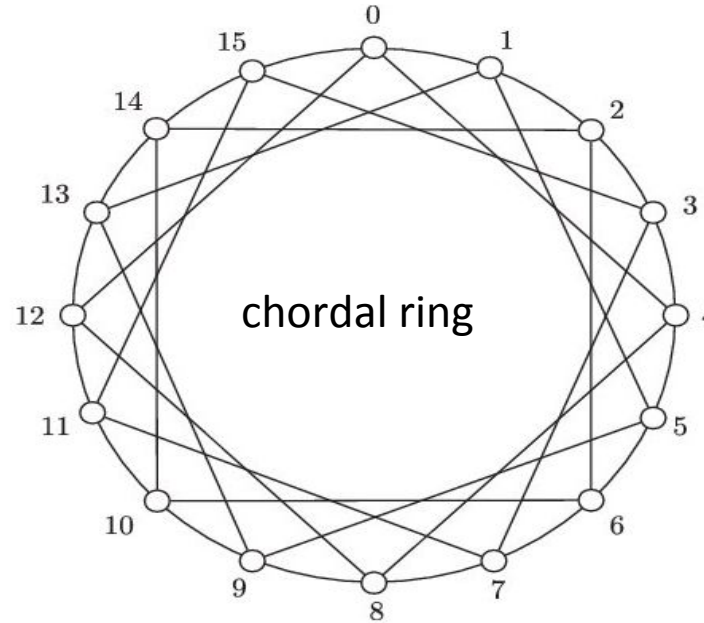
- In static network the interconnection network is fixed and permanent interconnection path between two processing elements and data communication has to follow a fixed route to reach the destination processing element.
- Thus, it Consist of a number of point to-point links.
- Topologies in the static networks can be classified according to the dimension required for layout i.e., It can be 1-D, 2-D, 3-D or hypercube.
- One dimensional topologies include Linear array as shown in figure.
- It is used in some pipeline architecture.



- Each node is connected to its two immediate neighboring nodes.
- If node i needs to communicate with node j , $j > i$, then the message from node i has to traverse nodes $i+1, i+2, \dots, j-1$.
- In the worst possible case, when node 1 has to send a message to node N , the message has to traverse a total of $N-1$ nodes before it can reach its destination.
- Linear arrays are simple in their architecture and have simple routing mechanisms.
- Linear arrays are slow. This is particularly true when the number of nodes N is large.

Ring and chordal ring Network

- If the two nodes at the extreme ends of a linear array network are connected, then the resultant network has ring (loop) architecture.
- By increasing the node degree from 2 to 3,4,...we obtain chordal rings.
- the more links are added, the higher the node degree and the shorter the Network diameter.



Barrel shifter network

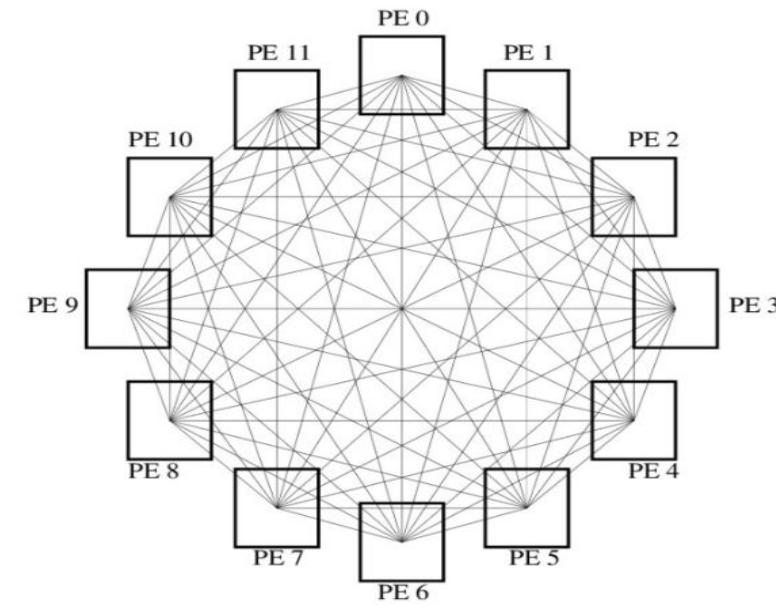
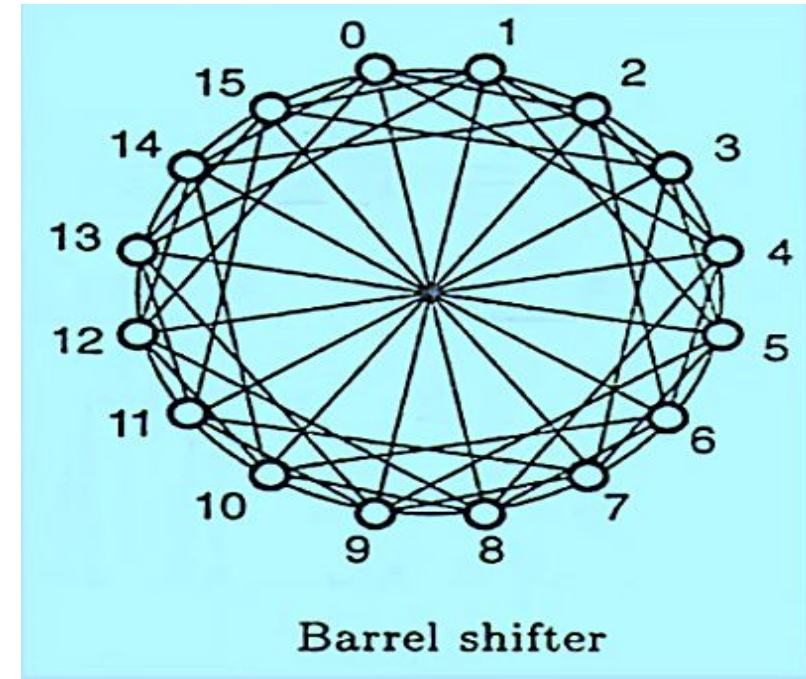
In a barrel shifter network the node diameter can be obtained by the formula:

$$D = n/2$$

$$\text{Node degree } d = 2n - 1$$

$$n = \log_2 N \text{ where } N \text{ is } 16$$

In completely connected network with processing elements 12
node degree is no. of PEs-1



Dynamic connection network

- Dynamic connection networks can implement all communication patterns based on program demands.
- In increasing order of cost and performance, these include
 - bus systems
 - multistage interconnection networks
 - crossbar switch networks.
- The price can be attributed to the cost of wires, switches, arbiters, and connectors
- Performance in this network is indicated by network bandwidth, data transfer rate, network latency, and communication patterns supported.

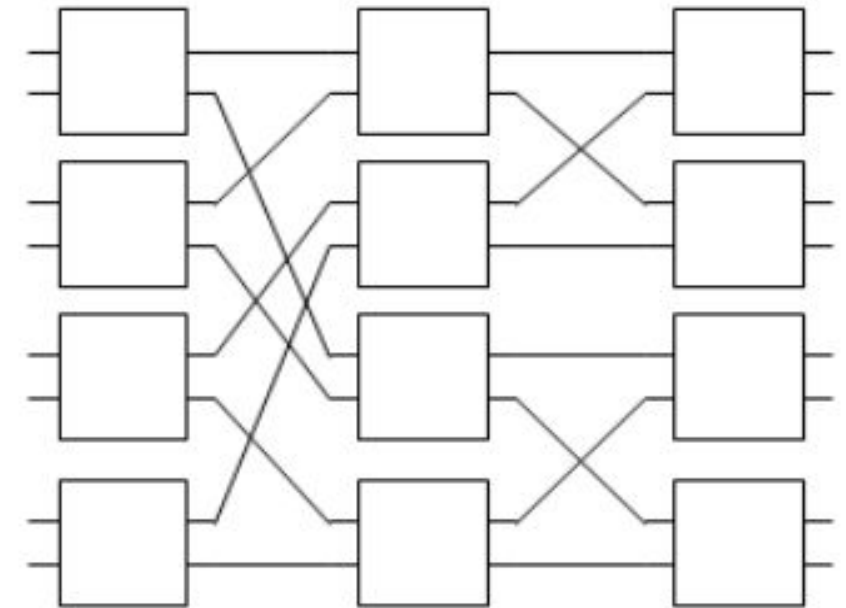
Bus systems

A bus system (contention bus, time-sharing bus) has

- a collection of wires and connectors
- multiple modules (processors, memories, peripherals, etc.) which connect to the wires
- data transactions between pairs of modules
- Bus supports only one transaction at a time.
- Bus arbitration logic must deal with conflicting requests.
- Lowest cost and bandwidth of all dynamic schemes.
- Many bus standards are available.

Switch modules

- An $a * b$ switch module has a inputs and b outputs. A binary switch has $a = b = 2$.
- It is not necessary for $a = b$, but usually $a = b = 2^k$, for some integer k .
- In general, any input can be connected to one or more of the outputs. However, multiple inputs may not be connected to the same output.
- When only one-to-one mappings are allowed, the switch is called a crossbar switch.
- The crossbar network provides highest bandwidth and interconnection capability.



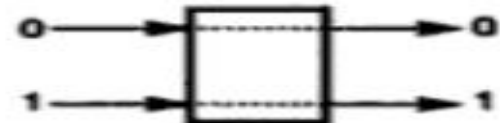
Crossbar network

Multistage network

- A generic multistage network consists of a sequence alternating dynamic switches (with relatively small values for a and b) with static networks (with larger numbers of inputs and outputs). The static networks are used to implement interstage connections (ISC).

Omega network

- A 2×2 switch can be configured for
 - Straight-through
 - Crossover
 - Upper broadcast (upper input to both outputs)
 - Lower broadcast (lower input to both outputs)
- With four stages of eight 2×2 switches, and a static perfect shuffle for each of the four interstage connections, a 16 by 16 Omega network can be constructed (but not all permutations are possible).
- In general , an n -input Omega network requires $\log_2 n$ stages of 2×2 switches and $n / 2$ switch modules.



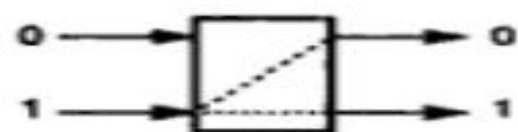
(a) Straight



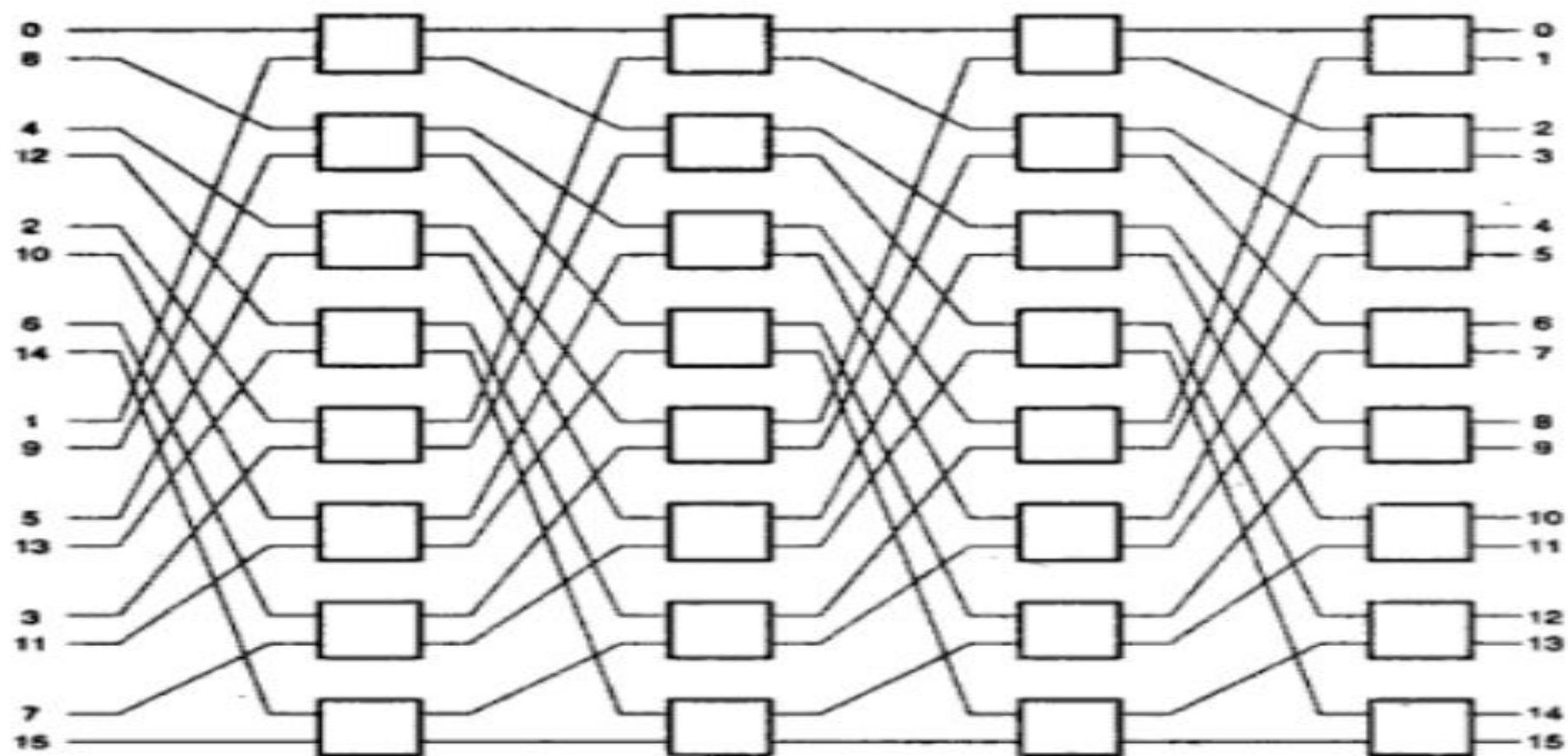
(b) Crossover



(c) Upper broadcast



(d) Lower broadcast



(e) A 16×16 Omega network

Omega Switching Network

How to Construct 8 x 8 OMEGA NETWORK

